



ST BENEDICT'S

SUBJECT	Science: Paper 2	DATE	19 July 2018
GRADE	11	MARKS	130
EXAMINER	Mrs Angelova	MODERATOR	Mr McCoy
NAME	MEMO	DURATION	120 min
CLASS			

QUESTION 1 – MULTIPLE CHOICE

12 MARKS

On the **answer sheet** make a cross (X) in the box corresponding to the letter which you consider to be the most correct.

1.1 C ✓✓

1.2 A ✓✓

1.3 A ✓✓

1.4 D ✓✓

1.5 A ✓✓

1.6 B ✓✓

2.1 The size/amount ✓ of the metal cube. ✓

Or The heat intensity of a flame ✓✓ should be kept the same. (2)

2.2 They only give the colour/presence of the flame ✓. No indication of ease of reactivity. ✓ Or the data is qualitative ✓, not quantitative ✓ (2)

2.3 Consider the reaction between lithium metal and oxygen to form lithium oxide.

2.3.1 Loss of electrons ✓✓ (2)

2.3.2 $\text{Li} \rightarrow \text{Li}^+ + 1\text{e}^-$ ✓ (2)

2.3.3 $2\text{Li} + \text{O} \rightarrow 2\text{Li}^+ + \text{O}^{2-}$ ✓ Balancing ✓ (3)

2.4.1 Free electrons move about the crystal ✓ Charge carrier ✓ (2)

2.4.2 Sodium contains metallic bonding ✓ Which is a strong intramolecular bond ✓

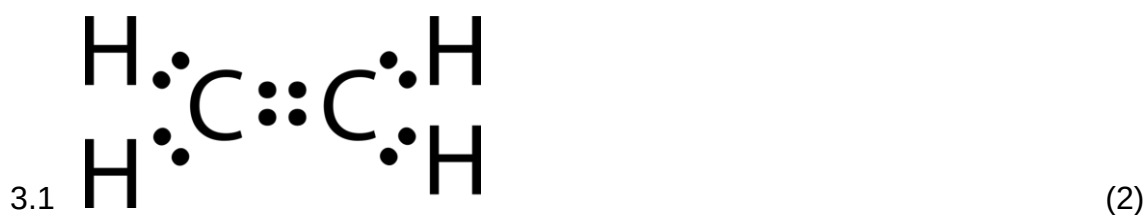
Oxygen is a discrete molecule with weak London IMFs ✓

Less energy needed to overcome the IMFs in oxygen compared to the intramolecular bond in Sodium ✓ (4)

2.5.1 $\text{K}_2\text{O} (\text{s}) \rightarrow 2\text{K}^+ (\text{aq}) + \text{O}^{2-} (\text{aq})$ ✓ (3)

2.5.2 Potassium ✓

Water is polar / Therefore has a positive and a negative region ✓
the negative region is oriented towards the ion ✓ therefore the ion would have to be positive (3)



3.3.a) $\text{END C} - \text{H} = 2,5 - 2,1 = 0,4 > 0 \checkmark$ Polar covalent bond $\checkmark$ (2)

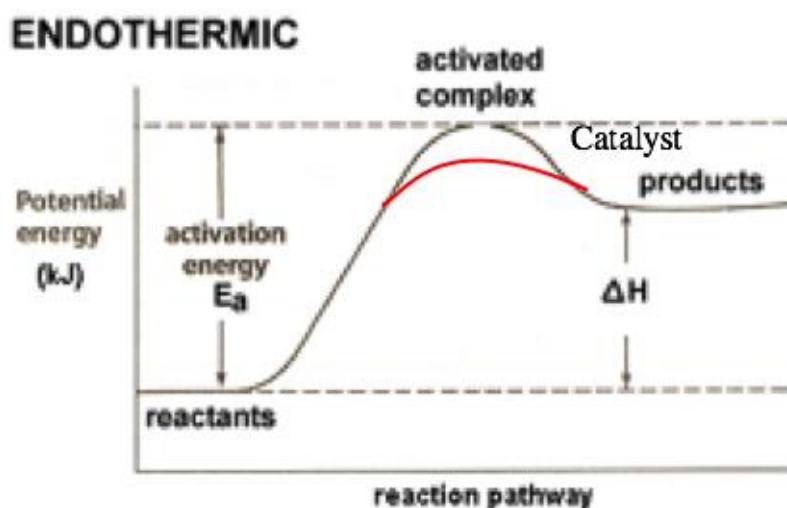
3.3.b) $\text{END C} = \text{C} = 0 \checkmark$ Pure covalent bond $\checkmark$ (2)

3.4 Electron cloud is symmetrical $\checkmark$

Non-polar molecule $\checkmark$ (3)

3.5.1 Activated complex: a high energy, unstable, temporary transition state between the reactants and the products $\checkmark\checkmark$ (2)

3.5.2



Labels of axis $\checkmark$

Shape $\checkmark$

Activated complex $\checkmark$

$\Delta H \checkmark$

Catalyst $\checkmark\checkmark$

3.5.3 substance that increases the rate of the reaction but remains unchanged at the end of the reaction $\checkmark\checkmark$ (2)

3.5.4 On diagram $\checkmark$ (1)

3.5.5 the net change of chemical potential energy of the system ✓✓ (2)

3.5.6 When the existing bonds are broken, energy is **Absorbed** ✓ (1)

3.5.7 when the new bonds are formed, energy is **released** ✓ (1)

3.5.8 the energy gained in the endothermic process is greater ✓ than the energy given out during the exothermic process ✓

OR

More energy is absorbed than released. ✓✓

OR

More energy is given out than taken in. ✓✓ (2)

QUESTION 4**(START ON A NEW PAGE)****27 MARKS**4.1 Al_2O_3 ✓✓

(2)

4.2.1 NaOH ; $n = cv$ H_2O : $n = m/M$

$$= 2 \times 0,2$$

$$= 12/18$$

$$= 0,4 \text{ moles available } \checkmark$$

$$= 0,67 \text{ moles available } \checkmark$$

 NaOH : H_2O

$$2 : 3$$

$$0,4 : 0,6 \quad \text{Enough} \quad \checkmark$$

0,44: 0,67 Not enough

0,67 moles of H_2O available, but only 0,6 required. Therefore NaOH is limiting ✓

(4)

4.2.2 measure of the extent of a reaction ✓ generally measured by comparing the amount of product against the amount of product that is possible ✓ (2)

4.2.3 Theoretical yield of $\text{NaAl}(\text{OH})_4$: NaOH : $\text{NaAl}(\text{OH})_4$

$$1: 1$$

0,4 moles of $\text{NaAl}(\text{OH})_4$ produced ✓Mass of $\text{NaAl}(\text{OH})_4$: $m = n \times M$ ✓

$$= 0,4 \times 118$$

$$= 47,2 \checkmark$$

$$\% \text{ yield} = \text{actual} / \text{theo} \times 100$$

$$= 45/47,2 \times 100 \checkmark$$

$$= \mathbf{95,33\%} \checkmark$$

(5)

4.3.1 Fe : $n = m/M$ Fe : Fe_2O_3

$$= 1000 \checkmark / 56$$

$$2: 1 \checkmark$$

$$= 17,86 \text{ mols produced } \checkmark$$

8,93 moles of Fe_2O_3 required (3)4.3.2 $N_{\text{Fe}_2\text{O}_3} = n \times N_A$ Fe : Fe_2O_3

$$= 8,93 \times 6,02 \times 10^{23} \checkmark$$

$$2: 1$$

$$= 5,38 \times 10^{24} \text{ Iron oxide units } \checkmark$$

$$2 \times 5,38 \times 10^{24} \checkmark = \mathbf{1,075 \times 10^{25} \text{ Fe atoms } \checkmark}$$

(4)

4.3.3 moles of CO : 26,79 moles ✓

$$V = nVm$$

$$= 26,79 \times 22,4 \checkmark$$

$$= \mathbf{600 \text{ dm}^3} \checkmark \text{ unit and value}$$

(3)

4.3.4 original mass of Fe_2O_3 required : $m = nM$

$$= 8,93 \times 160 \checkmark$$

$$= 1428,8 \text{ g } \checkmark$$

Mass $\times 70\% = 1428,8 \checkmark$ **Mass = 2041,14 g of Fe_2O_3 required** ✓**OR** $8,93 / 0,7 = 12,75$
moles of Fe_2O_3 Mass: $m = nM$

$$= 12,75 \times 160$$

$$= \mathbf{2041,14 \text{ g of } \text{Fe}_2\text{O}_3}$$

QUESTION 5**(START ON A NEW PAGE)****17 MARKS**

5.1 water that is associated with crystal formation ✓✓

OR . "water of crystallization" refers to water that is found in the crystalline framework of a salt. ✓ (2)

5.2 number of water molecules ✓ in the crystal per unit of the salt ✓ (2)

5.3 Standard solution ✓ (1)

5.4

$N = cV$ $= 0,2 \times 10$ $= 2 \text{ mol} \checkmark$ $V = n/c$ $= 2/2 \checkmark$ $= 1 \text{ dm}^3 - 200 \text{ cm}^3 = 800 \text{ cm}^3 \checkmark$	$c_1v_1 = c_2v_2 \checkmark$ $10 \times 200 = 2 \times v_2$ $V_2 = 1000 \text{ ml} \checkmark$ Therefore, 800 ml must be added (0.8 dm³) ✓
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(3)

5.5 HCl: $n = cv$

$$= 2 \times 0,0235$$

$= 0,047$ moles of HCl reacted ✓ **0,0235 moles of Na₂CO₃ reacts ✓ (3)**

Na₂CO₃ : HCl

1: 2 ✓

5.6 Mass of water released: $5 - 2,491 = 2,509 \text{ g} \checkmark$

Moles of water = m/M

$$= 2,509/18 \checkmark$$

$= 0,139$ moles ✓ (3)

5.7 Ratio of water to dehydrated sodium carbonate

H₂O : Na₂CO₃

$0,139 : 0,0235 \checkmark$

$5.93: 1 \checkmark$

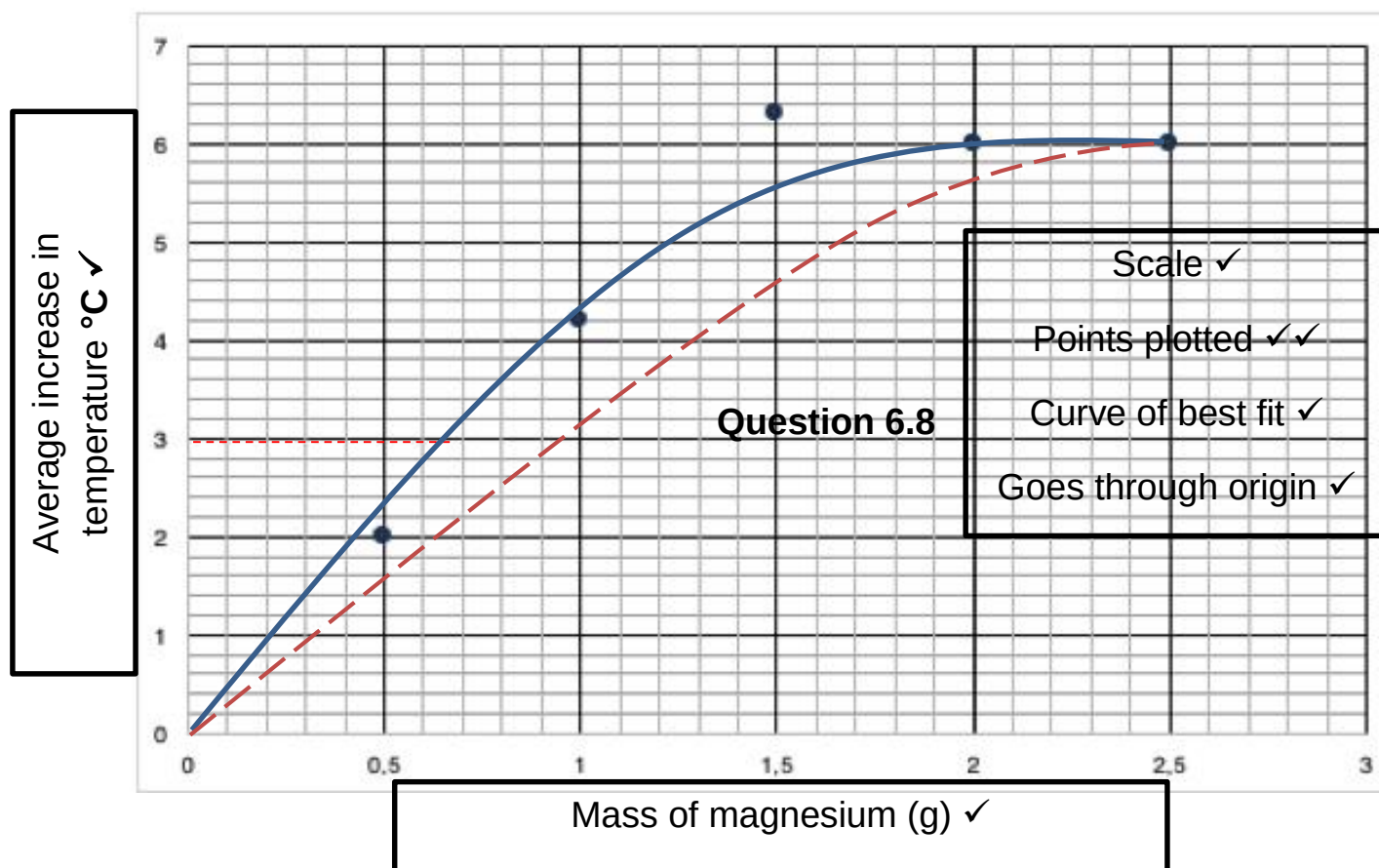
Therefore $x = 6 \checkmark$ (MUST be a whole number) (3)

QUESTION 6

(START ON A NEW PAGE)

26 MARKS

- 6.1 a measure ✓ of the average kinetic energy of the particles ✓ (2)
- 6.2 Increase/change ✓ in temperature ✓. (2)
- 6.3 NOT valid ✓ The results for the 2,0g mass are precise ✓ (2)
Or he cannot comment on accuracy as he has no known normal ✓
- 6.4 (7)



- 6.5 Answer sheet ✓ 0,65 – 0,7 g ✓
- 6.6 The copper sulfate solution *became* limiting ✓ No more magnesium could react ✓ (2)
- 6.7 Reaction was exothermic and thus producing heat ✓ OR The temperature increased as the rate of the reaction increased. ✓
The magnesium must have been limiting , therefore there are more reactant particles ✓ available so the chance of successful collisions every second increases ✓ as there will be more collisions with sufficient energy and correct orientation ✓. (4)
- 6.8 Less steep curve ✓ Ans sheet end at same point ✓ (2)
- 6.9 The surface area decreased ✓ Less exposed solid reactant particles. ✓ The chance of successful collisions every second decreases ✓ (3)